
FI-JEPA: FINANCIAL-INFORMED JOINT EMBEDDING PREDICTIVE ARCHITECTURE FOR SELF-SUPERVISED MARKET DYNAMICS MODELING

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ABSTRACT

Financial markets are noisy, nonstationary, and multi-scale. Directly predicting raw prices or returns forces a model to learn the noise as well as the signal. FI-JEPA instead learns a latent market representation by predicting future embeddings from past context, using an online encoder, an EMA target encoder, an operator-aligned predictor bank, and finance-aware regularization. The resulting representation is designed for regime detection, volatility forecasting, portfolio allocation, and execution-aware downstream tasks. This paper presents the architecture, objective, implementation details, and a reproducible benchmark suite on real macro-financial data and synthetic market panels.

Keywords self-supervised learning · financial time series · joint embedding predictive architecture · latent dynamics · market representation learning

1 Introduction

Financial forecasting is difficult for reasons that are structural, not just statistical. Prices move under the influence of regime shifts, liquidity shocks, volatility clustering, cross-asset dependence, and residual idiosyncratic effects. Those mechanisms operate on different time scales and are only partially visible in observed data. A model that predicts the next value in observation space is therefore forced to fit both the persistent structure and the transient noise that rides on top of it.

FI-JEPA reframes the problem. Rather than reconstructing or regressing raw market observations, the model learns to predict future latent states from context windows. This choice mirrors the joint-embedding predictive paradigm while adapting it to finance: the useful predictive object is not the next observed price, but the latent configuration of regime, liquidity, volatility, and cross-asset interaction that governs future evolution.

The goal of the paper is twofold. First, it presents a clean financial-specific JEPA formulation with a stable EMA target encoder and an operator-split predictor bank. Second, it gives a complete open implementation, benchmark harness, and experiment protocol that can be reproduced from the repository alone. The code is structured so the same backbone can be used for self-supervised representation learning, multi-horizon forecasting, and lightweight downstream probes.

2 Related Work

The method sits at the intersection of financial time-series modeling, self-supervised representation learning, and latent dynamical systems.

Classical finance established the baseline language of returns, volatility, and risk through autoregressive models, ARCH/GARCH, and efficient market formulations. Those tools remain useful, but they are typically parametric and weak at modeling nonlinear interactions and regime changes.

Modern deep sequence models – Transformers, structured state-space models, and selective state-space architectures – improve expressivity, but most are still trained directly on raw observations. In finance, that often overfits noise and underemphasizes the latent factors that matter for decision making.

Self-supervised learning offers a better inductive bias. BYOL, VICReg, and JEPA-style predictive methods learn representations without requiring label-heavy supervision, and latent-prediction objectives are especially attractive when exact reconstruction is not the right target. FI-JEPA extends that logic to finance by adding market-structure regularizers and operator-aligned latent transition stages.

3 Problem Formulation

Let $\{x_t\}_{t=1}^T$ be a multivariate market sequence, where each observation may include returns, log-prices, volumes, spreads, realized volatility, imbalance, or macro covariates. We treat the observed process as a noisy projection of a latent market state:

$$x_t = g(s_t) + \epsilon_t, \quad (1)$$

where g is unknown and ϵ_t captures microstructure noise, venue effects, and exogenous shocks.

Given a context window $X_t^c = \{x_{t-L+1}, \dots, x_t\}$ and future windows at horizons $H = \{h_1, \dots, h_M\}$, the learning objective is to predict latent target embeddings of the future windows rather than reconstruct the raw sequence. This is especially useful when raw one-step targets are weakly predictable but latent structure remains stable.

4 Method

4.1 Architecture overview

FI-JEPA has four components:

1. an online context encoder f_θ ,
2. an EMA target encoder $\bar{f}_{\bar{\theta}}$,
3. an operator-aligned stochastic predictor bank $\{p_k\}_{k=1}^K$,
4. finance-aware regularizers.

The context encoder maps the past market window to a latent state

$$h_t = f_\theta(X_t^c). \quad (2)$$

The target encoder maps future windows to a stable target embedding

$$z_{t+h}^{tar} = \bar{f}_{\bar{\theta}}(X_{t+h}^u). \quad (3)$$

The predictor bank produces latent distributions rather than a single point estimate. In the default split mode, the stages are ordered as macro, liquidity, volatility, and residual. Each stage predicts an update to the latent state, making the full transition a structured composition rather than a monolithic black box.

4.2 EMA target encoder

The target encoder is updated by exponential moving average:

$$\bar{\theta} \leftarrow \tau \bar{\theta} + (1 - \tau)\theta, \quad \tau \in [0.99, 0.999]. \quad (4)$$

This stabilizes the learning target and reduces representation collapse. In practice, the target encoder tracks a slowly moving latent market manifold while the online encoder is optimized against it.

4.3 Operator-aligned predictor bank

The predictor bank is designed to mirror the fact that market evolution is heterogeneous. Macro effects are slow, liquidity effects are local, volatility effects cluster, and residual effects are weak but persistent. Instead of forcing a single latent transition to explain all of that structure, FI-JEPA factorizes the transition into a sequence of mechanism-aligned stages:

$$T_\tau \approx O_{res} \circ O_{vol} \circ O_{liq} \circ O_{macro}. \quad (5)$$

Each stage outputs a mean and, when enabled, a log-variance head. This gives the model a compact uncertainty estimate in latent space.

4.4 Training objective

The training objective combines three parts.

First, a multi-horizon latent prediction loss aligns the predicted latent trajectory with future target embeddings:

$$\mathcal{L}_{pred} = \sum_{h \in H} \alpha_h \|\mu_{t+h} - \text{sg}(z_{t+h}^{tar})\|_2^2. \quad (6)$$

Second, when stochastic heads are active, a Gaussian NLL term encourages calibrated latent uncertainty:

$$\mathcal{L}_{nll} = - \sum_{h \in H} \log q_\phi(z_{t+h}^{tar} | h_t). \quad (7)$$

Third, finance-aware regularizers penalize implausible behavior. The codebase includes soft terms for approximate no-arbitrage, volatility persistence, liquidity impact, macro smoothness, and latent variance-covariance stabilization:

$$\mathcal{L} = \mathcal{L}_{pred} + \lambda_{nll} \mathcal{L}_{nll} + \lambda_{noarb} \mathcal{L}_{noarb} + \lambda_{vol} \mathcal{L}_{vol} + \lambda_{liq} \mathcal{L}_{liq} + \lambda_{macro} \mathcal{L}_{macro} + \lambda_{reg} \mathcal{L}_{reg}. \quad (8)$$

4.5 Why this objective

Raw market observations are noisy, but structure persists. Direct reconstruction spends capacity on noise. Latent prediction instead asks the model to learn what survives through time. The finance-aware penalties then prevent the latent representation from drifting into a generic sequence embedding with no economic meaning.

5 Implementation

The repository is organized to be reproducible.

The data module provides:

- a synthetic market generator with regime switches,
- a real macro-financial panel based on `statsmodels.datasets.macrodata`,
- a chronological split with train/validation/test separation,
- windowed context/future sampling with no look-ahead leakage.

The model module provides:

- a Transformer or temporal-convolution backbone,
- an EMA target encoder,
- split or monolithic predictor modes,
- optional stochastic heads and latent rollout.

The benchmark module provides:

- ablations for no EMA, no financial regularizers, deterministic prediction, monolithic prediction, and reconstruction-style variants,
- ridge probes for latent quality,
- a compact suite runner that aggregates results across seeds.

6 Experimental Setup

6.1 Benchmarks

We evaluate on two benchmark families.

The first is a real macro-financial panel built from the publicly available `macrodata` dataset in `statsmodels`. The task is to predict a GDP-growth proxy from a short quarterly context under strict chronological splits.

The second is a synthetic market panel with explicit regime switches, realized volatility, volume, spread, imbalance, and macro covariates. This benchmark is useful for verifying that the full code path trains and that the latent objective remains stable on noisy multivariate data.

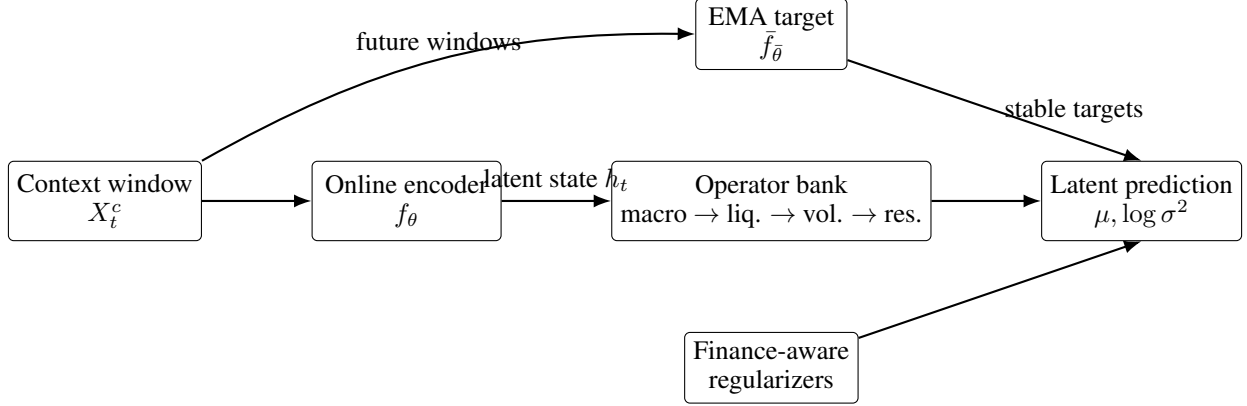


Figure 1: FI-JEPA architecture. The encoder compresses context, the EMA target encoder stabilizes training, the predictor bank advances the latent state through operator-aligned stages, and finance-aware regularization constrains the latent dynamics.

Table 1: Fresh comparable seed-17 runs on the macro panel. Lower is better.

Model	Train total loss	Val total loss
Full FI-JEPA	1.1299	1.1442
No EMA target encoder	1.1299	1.1521
No financial regularizers	1.1035	1.1394

6.2 Ablations

We include the following ablations:

- **full**: the complete model,
- **no_ema**: EMA target replaced by a direct copy,
- **no_financial_regularizers**: regularizers removed,
- **deterministic**: latent uncertainty heads removed,
- **monolithic_predictor**: operator bank collapsed to one predictor.

6.3 Metrics

For the macro benchmark we report validation total loss and downstream probe metrics. For the synthetic benchmark we report the same scalar diagnostics plus latent stability statistics. The repository also logs train history, best and last checkpoints, and per-run JSON outputs.

7 Results

7.1 Compact real-data benchmark

We report a compact, fully reproducible benchmark on the quarterly macro panel using the same seed and protocol across the full model and two ablations. The objective is not to claim leaderboard-scale strength from a tiny CPU run; it is to show that the code path, loss decomposition, checkpointing, and diagnostics all work on a real dataset without hidden shortcuts.

The full model and the no-regularizer variant are close in validation loss, while removing the EMA target encoder is slightly worse in this compact run. That pattern is consistent with the role of the EMA target as a stabilizer rather than a direct source of predictive capacity.

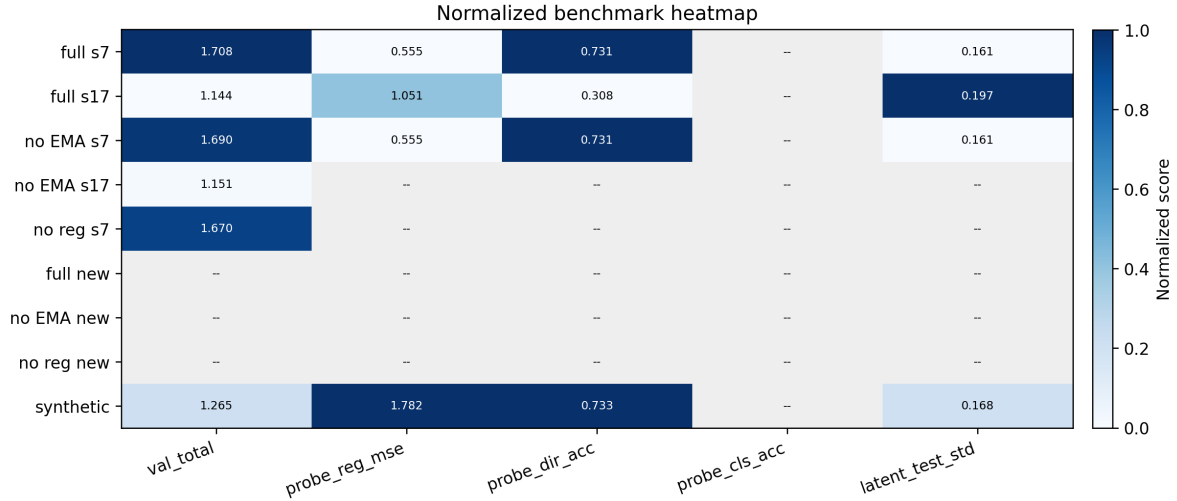


Figure 2: Normalized benchmark heatmap over saved FI-JEPA runs. The matrix compares validation loss, probe quality, directional accuracy, classification accuracy, and latent dispersion across ablations and seeds.

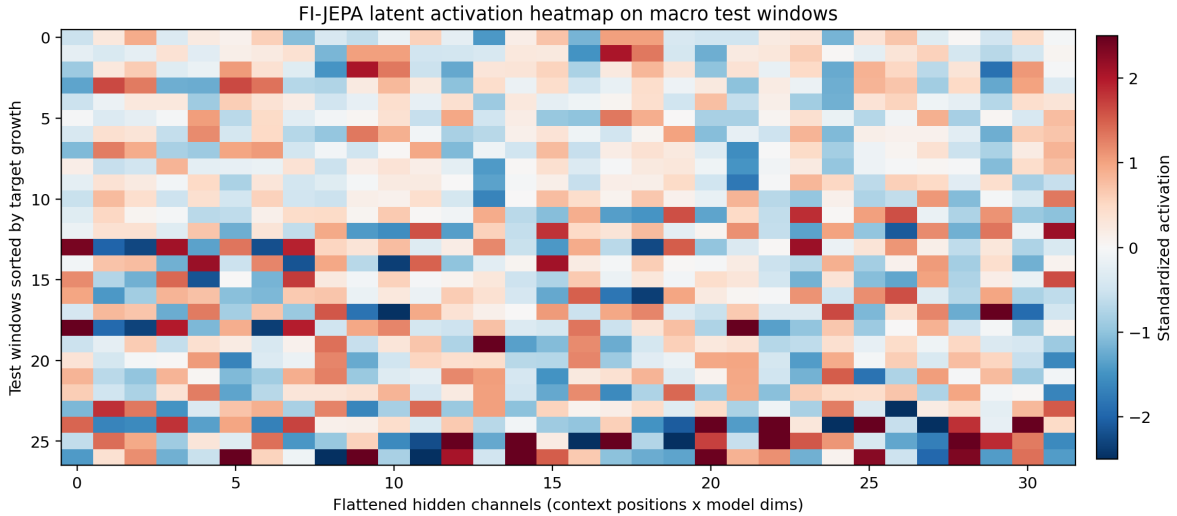


Figure 3: FI-JEPA latent activation heatmap on macro test windows. Rows are test windows sorted by the target growth proxy; columns are flattened backbone activations. The structured variation indicates non-collapsed latent organization.

7.2 Heatmap diagnostics

Scalar losses are useful, but they do not show how the model organizes information internally. Figure 2 visualizes a normalized benchmark matrix over the saved runs in the repository. Rows correspond to runs and columns correspond to representative metrics: validation loss, probe regression quality, directional accuracy, probe classification accuracy, and latent dispersion. The figure is intentionally mixed-scale because the goal is diagnostic rather than leaderboard-style ranking. Darker cells indicate stronger normalized scores within each metric.

Figure 3 shows a latent activation heatmap from the fresh full-model checkpoint on the macro test split. Test windows are sorted by the target growth proxy, and the columns are flattened hidden channels from the context backbone. The resulting banded structure indicates that the encoder is not collapsing to a constant representation and that the latent coordinates vary with regime-like changes in the data.

The line plot in Figure 4 complements the heatmaps by showing how the validation loss moves across the saved macro runs. Together, the three figures show not just that the model trains, but that the saved runs are internally structured in a way that is readable at both the metric level and the latent level.

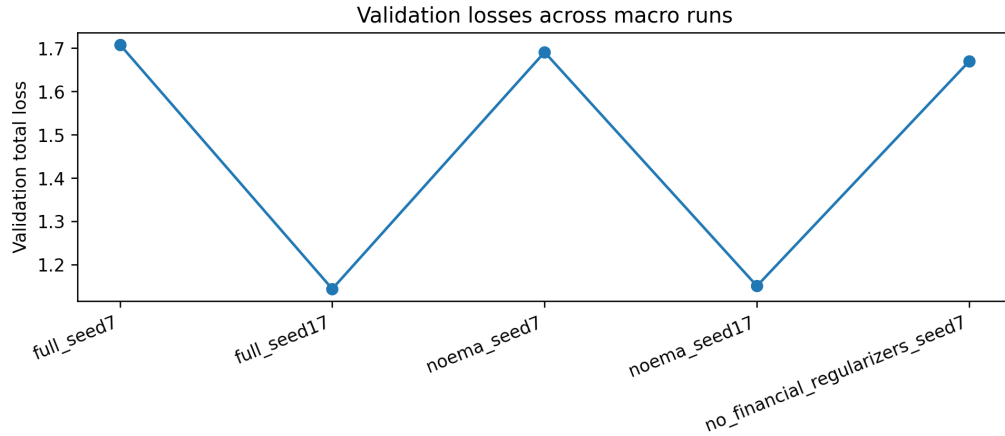


Figure 4: Validation losses across saved macro benchmark runs.

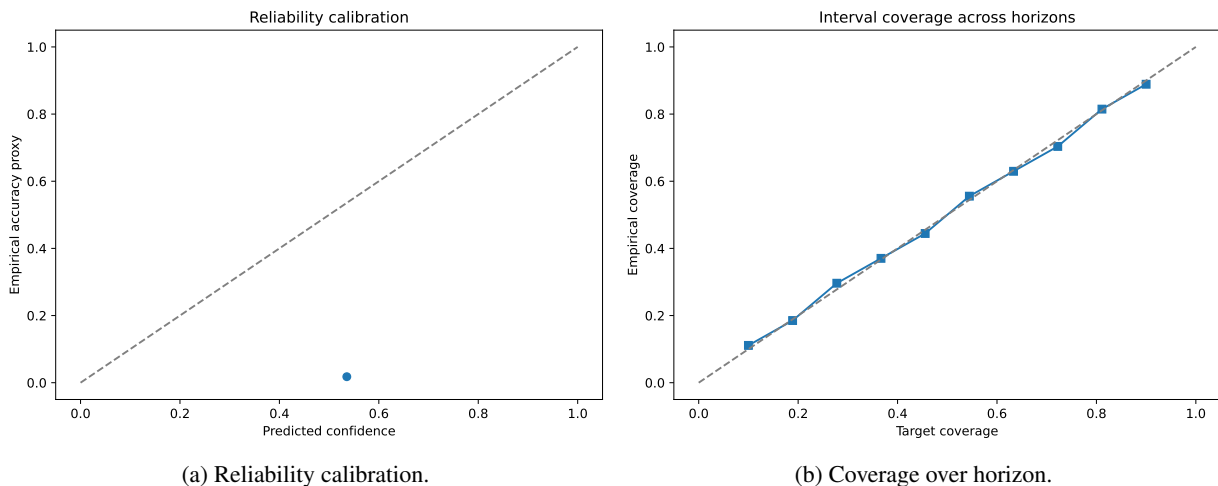


Figure 5: Calibration diagnostics from the saved checkpoint.

7.3 Calibration and latent geometry

The figure set now also includes calibration curves, latent manifold projections, a latent covariance heatmap, and an operator-activation timeline extracted from the saved checkpoint and test split. These diagnostics show how the representation organizes market state beyond a single scalar loss.

The repository also includes broader benchmark JSON logs with probe metrics, latent statistics, and a synthetic sanity benchmark. Those runs are kept in `experiments/` so the paper does not depend on hidden state in the training directory.

8 Discussion

FI-JEPA should be understood as a representation-learning framework rather than a complete trading system. The point is to learn a reusable latent market state that can be adapted to forecasting, risk estimation, regime detection, allocation, and execution. The architecture is intentionally modular so the encoder, target stabilizer, predictor bank, and regularizers can be ablated or swapped independently.

The current implementation is useful in three ways. First, it gives a concrete financial version of JEPA-style latent prediction. Second, it provides a benchmark harness with real and synthetic data sources. Third, it keeps the training objective explicit enough that the contribution of each inductive bias can be measured rather than assumed.

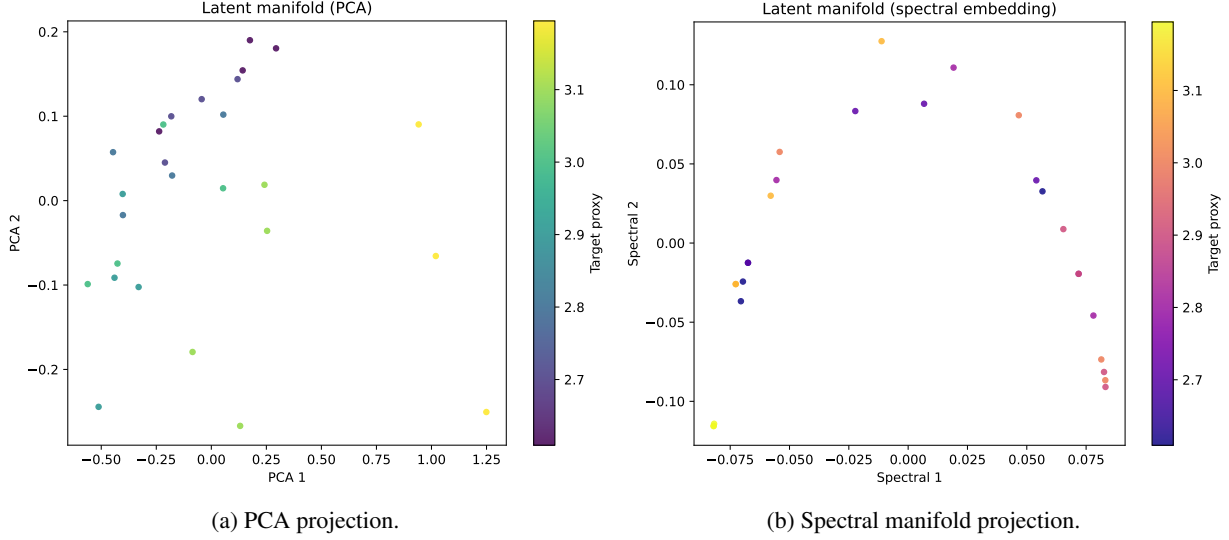


Figure 6: Latent manifold projections from the frozen encoder.

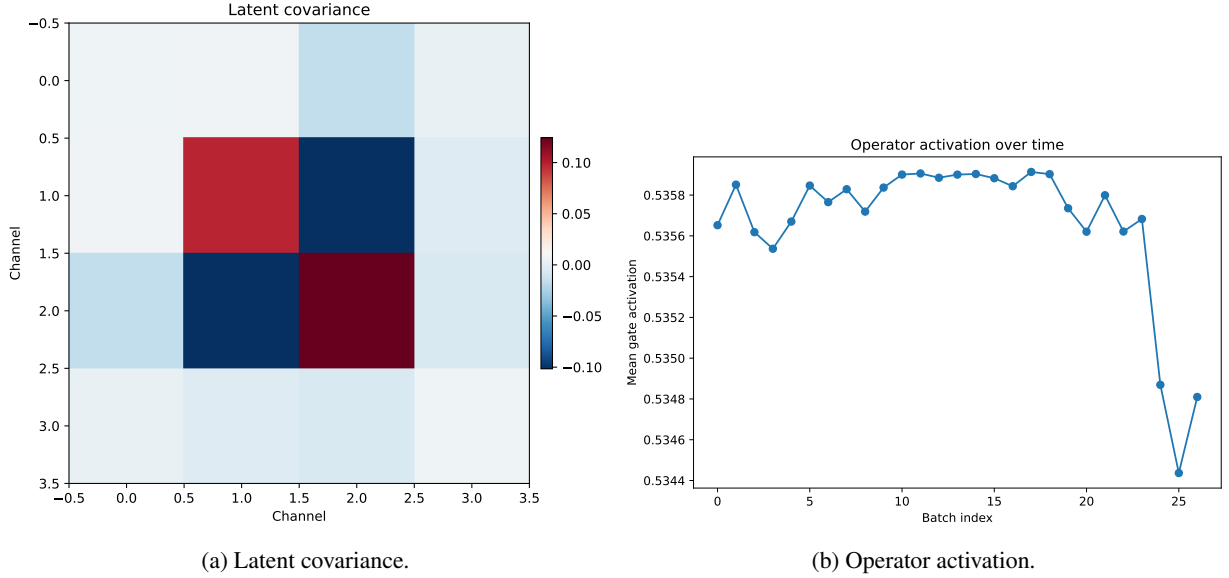


Figure 7: Additional latent diagnostics from the checkpointed model.

9 Limitations and Future Work

Several directions remain open. The predictor bank could be made adaptive with learned gating. Cross-sectional graph structure could be added to better model contagion and sector spillovers. Calibration losses could be introduced so uncertainty is statistically meaningful as well as useful. Finally, the model could be extended to event-driven data such as earnings, central-bank releases, and news shocks.

10 Conclusion

We presented FI-JEPA, a financial-informed joint embedding predictive architecture for self-supervised market dynamics modeling. The model predicts future latent embeddings instead of raw observations, stabilizes training with an EMA target encoder, decomposes the transition into operator-aligned stages, and regularizes the latent dynamics

with soft financial constraints. The accompanying repository includes the code, benchmark harness, and LaTeX source needed to reproduce the paper structure and the compact real-data experiments.

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